

Exam Date & Time: 22-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

Artificial Intelligence [CSE-4053]

Marks: 50

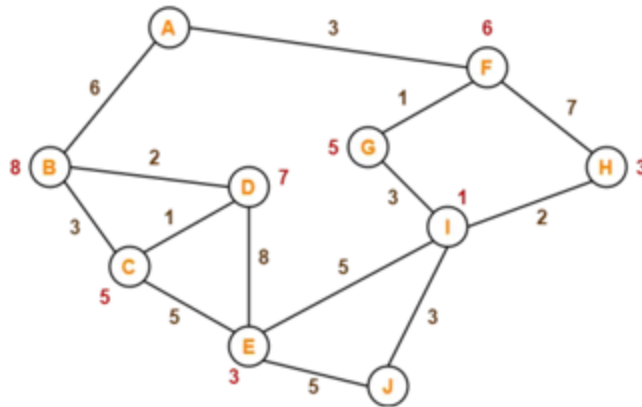
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze the limitations of the Turing Test as a measure of machine intelligence. (3)
 - A)
 - B) Apply the pivotal role of the Turing Test in the history of AI and show its importance. (3)
 - C) Explain the significance of Turing test in better understanding the evolution of AI. (4)
- 2) Analyse the significance of environment types in AI applications. (3)
 - A)
 - B) Design an intelligent agent architecture considering simple reflex and model based reflex agents and their respective environments for Medical Diagnosis System. (3)
 - C) Experiment PAGE for setting Interactive English Tutor with following parameters with values.
 - i) Performance measure: Student score on the test
 - ii) Environment: set of students and testing agency (4)
 - iii) Actuators: Display of exercises, Feedback, speech
 - iv) Sensors: keyboard entry, voice
- 3) Compare and contrast between uninformed search and informed search techniques along with at least one example from each. (3)
 - A)
 - B) (3)

Apply A* search on the following graph and find the path with cost for reaching from node A to node J.



- C) Do you agree that alpha-beta pruning is better over min-max strategy? Justify. (4)
- 4) The Wumpus world is a cave with 16 rooms (4×4). Each room is connected to others through walkways (no rooms are connected diagonally). The knowledge-based agent starts from Room [1, 1]. The cave has - some pits, a treasure and a beast named Wumpus. The Wumpus cannot move but eats the one who enters its room. If the agent enters the pit, it gets stuck there. The goal of the agent is to take the treasure and come out of the cave. The agent is rewarded when the goal conditions are met. The agent is penalized when it falls into a pit or being eaten by the Wumpus. Apply the PEAS specification for the wumpus world. (3)
- A) Define is resolution? Provide resolution for the following statements in Wumpus world. (3)
- i) There is a pit at 2,1 or 2,3 or 1,2 or 3,2.
- ii) Therefore (by resolution) the pit must be at 2,3 or 1,2 or 3,2
- B) Given a set of FOPL statements and a query about the domain, use logical inference rules or techniques to analyse and derive new conclusions. Consider FOPL statements and a query in the domain of mathematical operations. FOPL Statements: (4)
1. $\forall x, y (Add(x, y) \rightarrow GreaterThan(x, 0))$
 2. $\forall x, y (Multiply(x, y) \rightarrow Multiply(y, x))$
 3. $\forall x (GreaterThan(x, 0) \rightarrow GreaterThan(Square(x), 0))$
- Query:
"Is the product of any two positive numbers always positive?"
- 5) How predicate calculus is better than FOPL? Analyze. (3)
- A) Convert the following expression into clausal form; also, provide the detail steps with explanation. (3)
- B) $\exists x \exists w \forall y (\forall z P(f(x, y), w, z) \rightarrow (\exists u Q(w, u) \& \exists v R(y, v)))$

C) Present a detail analysis on forward reasoning and backward reasoning. (4)

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